Reaction Cross Sections of Excited Nuclei*

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The possible effect of excitation energy on the reaction cross section of Co^{59}, with protons and with α particles has been investigated. The investigation was effected by comparison of the measurements of the energy spectra of the α particles emitted in the $Ni^{62}(p,\alpha)Co^{59}$ and the protons emitted in the $Fe^{56}(\alpha,p)Co^{59}$ reactions. From the point-by-point ratio of the two spectra at kinetic energies that correspond to the same excitation energy of the residual Co^{59} nucleus, it is found that the ratio of inverse-reaction cross sections with protons and α particles does not differ significantly from that of reaction cross sections with protons and α particles determined with ground-state Co^{59} targets.

INTRODUCTION

THE statistical-model treatment of compound-nucleus reactions requires a knowledge of the inverse-reaction cross section. This quantity was introduced by Weisskopf[1] when he applied the principle of detailed balance to the description of compound-nuclear reactions. The inverse-reaction cross section at channel energy ϵ, $\sigma_{\rm inv}(\epsilon)$, refers to capture by nuclei in excited states rather than in the ground state, and it is an open question whether excited nuclei have the same capture cross section as ground-state nuclei. In a discussion of this topic, such arguments as lowering of the Coulomb barrier,[2,3] temperature dependence of optical-model parameters, and Pauli-principle restrictions[4] have been raised in order to show that $\sigma_{\rm inv}(\epsilon)$ can in general be different from $\sigma_R(\epsilon)$. It is only because of the lack of experimental or quantitative theoretical knowledge that in analyses of spectra it is generally assumed that these two quantities are identical.

We propose an experimental method for the investigation of a possible difference between $\sigma_{\rm inv}(\epsilon)$ and $\sigma_R(\epsilon)$. This method is based on the study of two or more nuclear reactions leading to the *same residual* nucleus such as the $Ni^{62}(p,\alpha)Co^{59}$ and $Fe^{56}(\alpha,p)Co^{59}$ reactions which we report on here.[5] In this situation the level density of the residual nucleus, $\rho(\epsilon_m-\epsilon)$, can be made the same in both reactions and thus will cancel when the point by point ratios of two evaporation spectra, $N_\alpha(\epsilon_\alpha)$ and $N_p(\epsilon_p)$, are considered at the same excitation energies $(\epsilon_m-\epsilon)$ of the residual nuclei. Specifically, the evaporation spectrum can be written in the following form:

$$N(\epsilon)=\text{const}\times\rho(\epsilon_m-\epsilon)\epsilon\sigma_{\rm inv}(\epsilon)f(\epsilon)\,, \qquad (1)$$

where $\rho(\epsilon_m-\epsilon)$ is the level density, ϵ is the channel energy, and $f(\epsilon)$ is a function which contains all factors other than energetically available phase space and inverse-reaction cross section affecting the shape of the energy spectrum. In this paper we shall incorporate only angular-momentum effects into $f(\epsilon)$. Applying Eq. (1) to each of the reactions [Ni^{62}, $(p,\alpha)Co^{59}$, and Fe^{56}-$(\alpha,p)Co^{59}$] and considering a common excitation energy U of the residual nuclei,

$$U=(\epsilon_{\max}-\epsilon)_\alpha=(\epsilon_{\max}-\epsilon)_p\,, \qquad (2)$$

it follows that

$$\left[\frac{\sigma_{\rm inv}(\epsilon_\alpha)}{\sigma_{\rm inv}(\epsilon_p)}\right]_U=\text{const}\times\left[\frac{\epsilon_pN_\alpha(\epsilon_\alpha)}{\epsilon_\alpha N_p(\epsilon_p)}\right]\frac{f_p(\epsilon_p)}{f_\alpha(\epsilon_\alpha)}\,. \qquad (3)$$

In the past, many analyses of experimental results based on Eq. (1) have been carried out without considering effects of angular-momentum conservation. If one does this, by assuming that the corrective term, $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$ is constant, then, as we can see from Eq. (3), the ratio of inverse-reaction cross sections becomes a function of quantities which can be deduced directly from experimental results and compared to the ratio of corresponding reaction cross sections with ground-state nuclei which are also experimentally measurable quantities. Any difference in shape between $[N_\alpha(\epsilon_\alpha)\epsilon_p/N_p(\epsilon_p)\epsilon_\alpha]_U$ and the corresponding ratio $[\sigma_R(\epsilon_\alpha)/\sigma_R(\epsilon_p)]$ would in this case be considered as an experimental indication that $\sigma_{\rm inv}(\epsilon)$ is not identical to $\sigma_R(\epsilon)$. In an analysis that includes the influence of spin distributions of the compound nuclei on the shape of evaporation spectra, additional investigation of the energy-dependent corrective term $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$ is necessary, and will be done later in this paper.

EXPERIMENTAL RESULTS

The reactions $Ni^{62}(p,\alpha)Co^{59}$ and $Fe^{56}(\alpha,p)Co^{59}$ were studied at bombarding energies 14.3 and 18.2 MeV, respectively, with the Columbia University cyclotron and the Yale University heavy-ion linear accelerator. Energy and angular distributions of the emitted particles were measured with a semiconductor counter telescope placed in a 1-m-diam scattering chamber.

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TABLE I. Reaction cross sections of α particles and protons with Co^{59} as a function of channel energy.[a]

channel energy ϵ_α (MeV)	6.5	7.0	7.5	8.0	8.5	9.0	9.5	10.5
$\sigma_R(\epsilon_\alpha)$ (mb)	4.2	12	33.8	75.5	130	200	320	440
channel energy ϵ_p (MeV)	5.5	6.0	6.5	7.0	7.5	8.0	8.5	10
$\sigma_R(\epsilon_p)$ (mb)	290	350	430	490	555	615	690	810

[a] The data are from Refs. 6–9.

Typical backward spectra in the center-of-mass system are shown in Fig. 1, where the channel energies ϵ_α and ϵ_p have been matched to give the same excitation energy ($U=14.5-\epsilon_\alpha=13.9-\epsilon_p$) of the residual Co^{59} nuclei. The angular distributions of emitted particles leading to residual excitation energies between 4.5 to 8.0 MeV are symmetrical about 90°, and, indeed, are very nearly isotropic. In addition, this region of both spectra has identical shapes at all angles from 30° to 150°, and thus the differential cross sections rather than energy spectra that are integrated over angle could be used in Eq. (3). The ratio $\sigma_{\rm inv}(\epsilon_\alpha)/\sigma_{\rm inv}(\epsilon_p)$ as a function of excitation energy derived in this manner from the spectra in Fig. 1 is shown in Fig. 2.

Some experimental values for the ground-state reaction cross sections σ_R in the $Co^{59}+\alpha$ and $Co^{59}+p$ systems are available for comparison with the results presented in Fig. 2. For $Co^{59}+p$, the reaction cross sections at energies above 8 MeV have been measured by the attenuation technique,[6] while those at lower energies have been measured as the sum of the cross sections for the $Co^{59}(p,n)$ and $Co^{59}(p$, charged particles) reaction.[7] The reaction cross sections of α particles with Co^{59} at several energies above 10 MeV were determined in this laboratory[8] by summing measurements of the cross sections for the $Co^{59}(\alpha,n)$ and $Co^{59}(\alpha$, charged particle) reactions; these results joined smoothly with those reported by Stelson, *et al.*,[9] at lower energies. Table I presents a summary of the reaction cross section used for purposes of comparison in this investigation.

Ratios of the ground-state reaction cross sections of Co^{59} with α particles to those with protons at the corresponding kinetic energies required by Eq. (2) are also shown in Fig. 2, where they are compared with the corresponding ratios determined from the spectra in Fig. 1 through use of Eq. (3) ignoring any possible energy dependence of $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$. The two ratio curves were *normalized to the same value* at the lowest residual excitation energy (highest ϵ_α) considered in this analysis because at that point any possible effect that would cause them to diverge should be minimal. The errors that are shown in Fig. 2 arise from those in

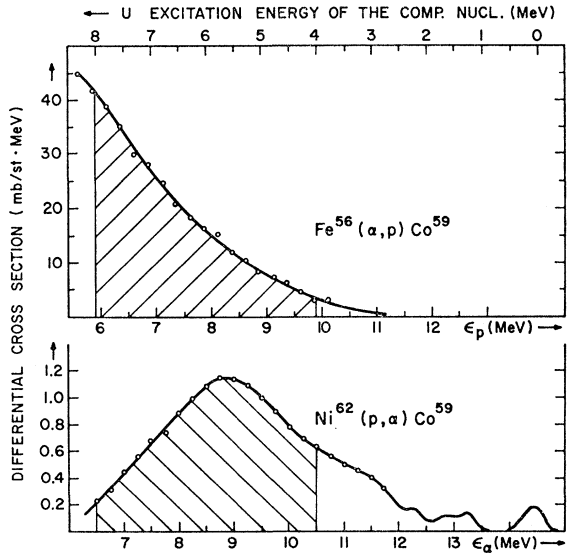


FIG. 1. Energy spectra of products of $Ni^{62}(p,\alpha)Co^{59}$ and $Fe^{56}(\alpha,p)Co^{59}$ reactions at the laboratory angle of 135°. The shaded areas were analyzed in this work.

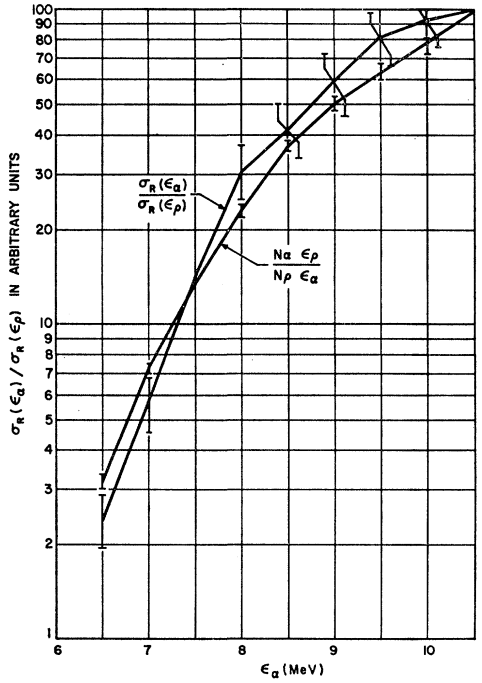


FIG. 2. The ratio $\sigma_{\rm inv}(\epsilon_\alpha)/\sigma_{\rm inv}(\epsilon_p)$ calculated with Eq. (3) from the data given in Fig. 1 compared with the ratio $\sigma_R(\epsilon_\alpha)/\sigma_R(\epsilon_p)$ taken from the data given in Table I. The ratios are normalized to the same value at the highest energy point.

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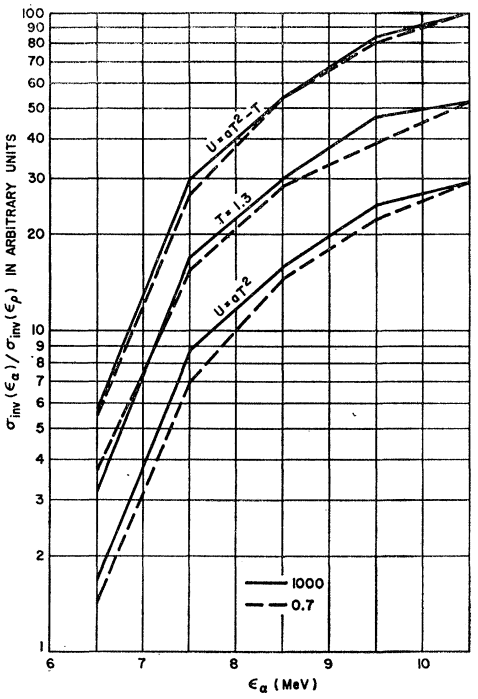


FIG. 3. The ratio $\sigma_{inv}(\epsilon_\alpha)/\sigma_{inv}(\epsilon_p)$ calculated with Eq. (3) from computer-generated spectra. The three sets of curves correspond to the assumptions about the energy dependence of the nuclear temperature that are given in the figure. The two curves in each set correspond to K=1000 and K=0.7.

the individual cross sections that enter into the computation of the ratio. The standard deviation of each ground state σ_R was taken to be $\pm 15\%$, while that for $\epsilon_p N_\alpha(\epsilon_\alpha)/\epsilon_\alpha N_p(\epsilon_p)$ was taken from the counting statistics in the experiments that gave the curves in Fig. 1. Additional uncertainties in the curves given in Fig. 2 which are possibly of comparable magnitude to those shown, but considerably more difficult to estimate, arise from uncertainties in the absolute energy-scale of the spectra shown in Fig. 1 and the σ_R given in Table I as well as the effect of the experimental energy resolution on the shapes of the spectra shown in Fig. 1. The effects of these *uncertainties* are magnified because the interesting quantities are the *ratios* of cross sections at *corresponding* energies.

DISCUSSION

On the basis of the simplified analysis in which the energy dependence of the $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$ corrective term is neglected, we would conclude from the curves in Fig. 2 that there is no evidence of a difference outside of experimental uncertainty between the ratios of σ_R for α particles and protons and that for σ_{inv}. Before this conclusion is taken seriously, however, it is necessary to investigate whether the lack of significant divergence between the two curves in Fig. 2 merely reflects the compensation of an excitation-energy effect on σ_{inv} by the energy dependence of $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$ that is imposed by angular-momentum conservation. This investigation has been carried out with the aid of the computer program EVAMCO[10] which calculates the energy spectrum of emitted particles by the usual statistical-model formalism including angular-momentum conservation.[11]

In order to find the correction term the calculations were carried out both with a very large moment of inertia for the residual Co^{59} nuclei and with a more realistic value. According to Eq. (1), the correction term would simply be the ratio of the two computed results because a very large moment of inertia gives the same spectrum that would be obtained without imposing angular-momentum conservation. The transmission coefficients required for this calculation were, of course, taken to be independent of the excitation energy of the residual nuclei. They were obtained from the ABACUS,-2 program[12] using optical-model parameters which are the same as those given by Hodgson[13] except for r_0 which was changed from 1.7 and 1.2 to 1.5 and 1.12 F for α particle and protons, respectively, in order to more faithfully reproduce the ground-state cross sections given in Table I. The spin-dependent level density was taken as

$$\Omega(U,J) = \frac{\exp[2a^{1/2}(U-\delta)^{1/2}]\exp[-J(J+1)/2\sigma^2]}{(U-\delta)^{5/4}\pi^{1/2}(2\sigma^2)^{3/2}}(2J+1),$$

with $a=A/11$ and pairing energies δ as given by Cameron.[14] The spin cutoff parameter σ^2 was taken to be

$$\sigma^2=(IT/\hbar^2)=0.0142\,KTA^{5/3},$$

where T is the nuclear temperature in MeV, I is the nuclear moment of inertia, and K is the ratio of I to the moment of inertia of a rigid sphere with mass number A and radius $R=1.2A^{1/3}\times 10^{-13}$ cm. The straightforward way of evaluating the effect of angular-moment conservation on the ratio curves shown in Fig. 2 is simply to treat the computer-generated spectra in precisely the same manner as the experimental spectra given in Fig. 1. The results of doing this are illustrated in Fig. 3, where two ratio curves for $K=0.7$ and $K=1000$ and for three different dependencies of temperature on excitation energy are again normalized at the same ϵ_α as in Fig. 2. The ratio curve for $K=1000$ should be, and is, identical to the corresponding ratio curve of the reaction cross section calculated with ABACUS,-2 using the identical optical-model parameter as those used in the calculation of the spectra. The

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values of $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$, which are given by the ratio of points on the $K=1000$ curve to the corresponding points for $K=0.7$, are seen to be of the order of unity but to depend in detail on the form chosen for the dependence of T on U. Thus, this estimate of the effects due to angular-momentum conservation leaves intact the conclusion that, within experimental error, there is no significant effect of excitation energy on the ratio of reaction cross sections of Co^{59} with proton and with α particles.

It is of interest to note that when it is assumed that the temperature is constant at 1.3 MeV, a reasonable assumption for the excitation energies considered here, the resultant dependence of $f_p(\epsilon_p)/f_\alpha(\epsilon_\alpha)$ on ϵ_α is just of the right form to remove any divergence that may exist between the experimental curves shown in Fig. 2.

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Neutron Cross Sections of Calcium and Potassium from 2 to 8 MeV*

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Differential cross sections for elastic scattering of neutrons have been measured for calcium at neutron energies of 2.06, 3.29, 5.30, 5.88, 6.52, and 7.91 MeV, and for potassium at neutron energies of 2.06, 3.74, 4.33, 6.52, and 7.91 MeV. The measurements were made using time-of-flight techniques and a liquid-scintillation detector equipped with γ-ray discrimination. Measurements of total cross sections for both nuclei were made between 1.8 and 8.3 MeV. The experimental data were compared to optical-model calculations using the computer code ABACUS-2. Parameter-search procedures produced good fits to the cross sections. The potentials implied for the two nuclei are similar.

INTRODUCTION

THERE have been several recent studies of the scattering of fast neutrons by ^{39}K and ^{40}Ca. The γ-ray de-excitation following fast neutron bombardment has been observed for ^{40}Ca by Day[1] and Tucker[2] and for ^{39}K by Nichols *et al.*[3] The purpose of these investigations was to determine the properties of the low-lying states of ^{40}Ca and ^{39}K, respectively. The neutron elastic differential cross sections of ^{39}K have been measured at 1.49, 2.38, 3.00, and 3.76 MeV by Towle and Gilboy[4] and at 3.7 MeV by Kent *et al.*[5] Towle and Gilboy have also measured the inelastic cross sections to the first three levels of ^{39}K. For ^{40}Ca above 2 MeV the elastic differential cross sections have been measured at 2.0, 3.5, and 6.0 MeV by Seagrave *et al.*,[6] at 3.2 MeV by Becker *et al.*,[7] at 3.7 MeV by Kent *et al.*,[5] and at 4.1 MeV by Vincent *et al.*[6] Recent neutron total cross-section measurements of potassium and calcium have been made by Glasgow and Foster[8] from 3–15 MeV and by Stüwer *et al.*[9] from 4–6 MeV. The measurements of Stüwer *et al.* give values approximately 6% lower than those of Glasgow and Foster.

This study presents precision measurements of the elastic differential cross sections and total cross sections of calcium and potassium between neutron energies of 2 and 8 MeV. This information is useful because of the existing discrepancies in the total cross sections and because it extends the range of information on the neutron elastic scattering by these two nuclei to 7.9 MeV. It the cross sections are represented as scattering by a complex potential, elastic-scattering data over a relatively large energy range is required to determine the energy dependence of the potential parameters. As a part of this study an attempt was made to fit both the calcium and potassium data with optical-model parameters at each energy and to determine the energy dependence of the parameters.

EXPERIMENTAL ARRANGEMENT

The data were taken using the 6-MeV Van de Graaff accelerator of the University of Kentucky. Neutrons of energies up to 5.3 MeV were produced by the $T(p,n)^3He$ reaction and from 5.3–8.3 MeV by the $D(d,n)^3He$

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